BLY 121: GENERAL SURVEY OF HIGHER PLANTS

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The term higher plants is used for the group of plants that belong to the Division Spermatophyta. These are plants characterized by the production of seeds. The spermatophytes are the most highly developed of all plants, they are also the most highly evolved of all plants. Two main groups of plants, each belonging to a separate Sub - division of Spermatophyta, are classed as higher plants. These are the Gymnosperms which belong to the Sub – division Gymnosperma and the Angiosperms which belong to the Sub- division Angiosperma.

They both share the characteristics of the higher plants.

Characteristics of the Higher Plants.

1.The spermatophytes produce seeds. The seed is a special structure employed for the reproduction of these plants.

2.They exhibit alternation of generation, in which it is characterized by the alternation of two generations of gametophyte and sporophyte within the life cycle

3.They are characterized by the presence of trees and shrubs among many of the members.

4.Spermatophytes are not dependent on water for their sexual reproduction because they possess pollen tubes, a structure that transport structures that produce the male gamete to that which bears the female gametes.

5.Spermatophytes are characterized by a phenomenon known as pollination, a process in their sexual reproduction.

In spite of the apparent similarities between the gymnosperms and the angiosperms, this does not mean that there are no differences between them. This difference is also of importance in the relative success of the two groups of higher plants, it enable biologists to rank the Angiosperms as being more successful than the Gymnosperms.

The Gymnosperms

The gymnosperms constitute a major subdivision within the Division Spermatophyta. The Gymnosperms are however less highly developed of these two main Sub – divisions of the Division Spermatophyta. They are also not as widely known as the Angiosperms, which are also commonly known as the flowering plants.
The name Gymnosperm is derived from two Greek words, “gymnos” which means “naked” and “sperma” which means “seed”. The name gymnosperm can therefore be interpreted to mean seed-bearing plants which produce their seeds in the naked form, that is, without being

enclosed within pericarps are characteristic of seeds of the flowering plants which are usually produced as fruits in which the seeds are enclosed by fruit walls called pericarps.

Of the three main classes within the group of plants known as gymnosperms, only two are very popular. These are the cycads and the conifers.

The cycads survive as a few species of tropical palm-like trees, including one which is native to the USA, *Zamia pumila* the cardboard palm. This is found on sandy soils in Florida and is sometimes grown as a foliage plant. *Cycas* species are larger and are often used as ornamentals in tropical areas. The cycads can be viewed as beneficial as they form symbiotic associations with nitrogen fixing bacteria, but they have also been the subject of extermination programs since they are highly toxic to livestock. Examples of the cycad include *Cycas revolute* and *Zamia pumila.*

Conifers are often large and can dominate the plant life in some ecosystems because their stems continue to expand in width as well as length throughout the life of the plant. The older parts of the stem become woody, which provides a further distinction from the seedless vascular plants of which there are no surviving woody representatives. Conifer leaves are needle or scale-like.

Conifers are important economically as a tree crop for pulp and timber. Their ability to grow in areas that are unsuitable for other crop production is an asset for this purpose. Similarly, since most (though not all) conifers are evergreen they are valued as landscape plants. The evergreen habit does have its disadvantages since premature leaf death caused by pollution, disease or insect attack can be more damaging than in plants which produce a complete new flush of leaves each spring. Example of conifer includes the pine tree.

Characteristics of Gymnosperms

1. Gymnosperms are heterosporous.
2. They produce true seeds like other seed plants.
3. Gymnosperms are mostly trees.
4. Most species of gymnosperms are evergreen.
5. Their sporophytes are large, autotrophic and more complex than their gametophytes which are relatively very small and are dependent in their nutrition upon the sporophytes.
6. Reproduction in the gymnosperms does not depend on water because the transportation of their pollen to the ovules is by wind pollination followed by subsequent growth of the pollen tubes into ovules which contain the eggs.
7. The female gametophytes of gymnosperms bear eggs in very small archegonia.

Economic Importance of Gymnosperms

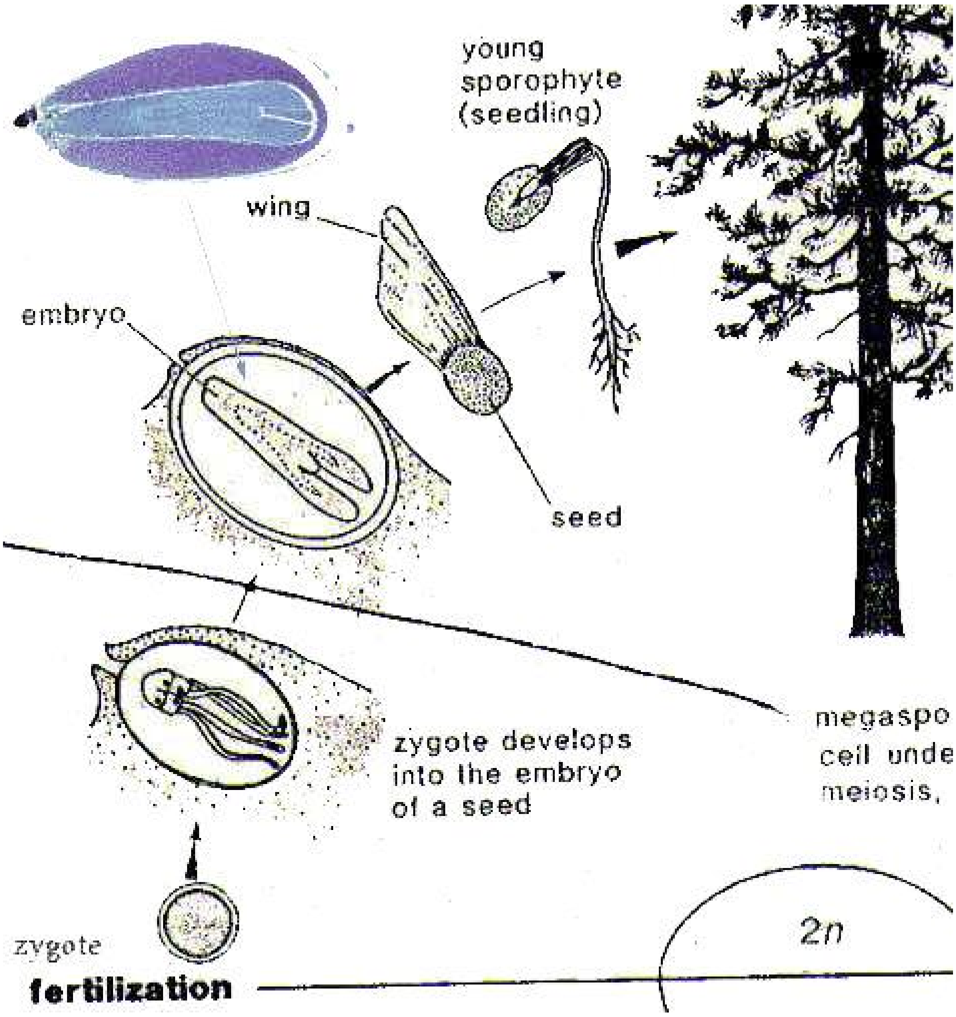
The gymnosperms are relatively small as a group of higher plants. Only a little over 700 species have been recorded. The group is important for the economy of man. Details of the role are as follows.

1. The conifers are important in many parts of the world as natural forest trees and they help in providing an effective cover over the land of the places they occupy and by so doing help in checking soil erosion, provides timber, shelter and also provides fuel.
2. The conifers also help in supplying the world with important soft wood such as pine, cedar, spruce, red wood and firs.
3. The conifers are source of many oils, resins, tars and turpentines.

4. Many species of gymnosperms such as Pinus, Encephalartos, Juniperus and Thuja are used as ornamentals in gardens and parks
5. The woods of conifers also help in providing wood gas, wood tar and wood alcohol.

Life cycle

Gymnosperms are spore-bearing plants (sporophytes), with a sporophyte-dominant life cycle; as in all other vascular plants the gametophyte (gamete-bearing phase) is relatively short-lived. Two spore types, microspores and megaspores, are generally produced in pollen cones or ovulate cones, respectively. A short-lived multicellular haploid, gamete-bearing phase (gametophyte) develops inside the spore wall. Pollen grains (microgametophytes) mature from microspores, and ultimately produce sperm cells; megagametophyte tissue develops in the megaspore of each ovule, and produces multiple egg cells. Thus, megaspores are enclosed in ovules (unfertilized seeds) and give rise to megagametophytes and ultimately to egg cells. During pollination, pollen grains are physically transferred between plants, from pollen cone to the ovule, being transferred by wind or insects. Whole grains enter each ovule through a microscopic gap in the ovule coat (integument) called the micropyle. The pollen grains mature further inside the ovule and produce sperm cells. Two main modes of fertilization are found in gymnosperms. Cycads and *Ginkgo* have motile sperm that swim directly to the egg inside the ovule, while conifers and gnetophytes have sperm with no flagella that are conveyed to the egg along a pollen tube. After fertilization (joining of the sperm and egg cell), the zygote develops into an embryo (young sporophyte). More than one embryo is usually initiated in each gymnosperm seed. Competition between the embryos for nutritional resources within polyembryonic seeds produces programmed cell death to all but one embryo. The mature seed comprises the embryo and the remains of the female gametophyte, which serves as a food supply, and the seed coat (integument). The following illustrations clearly depict or show the life cycle of gymnosperms.


young
sporophyte
(seedling)
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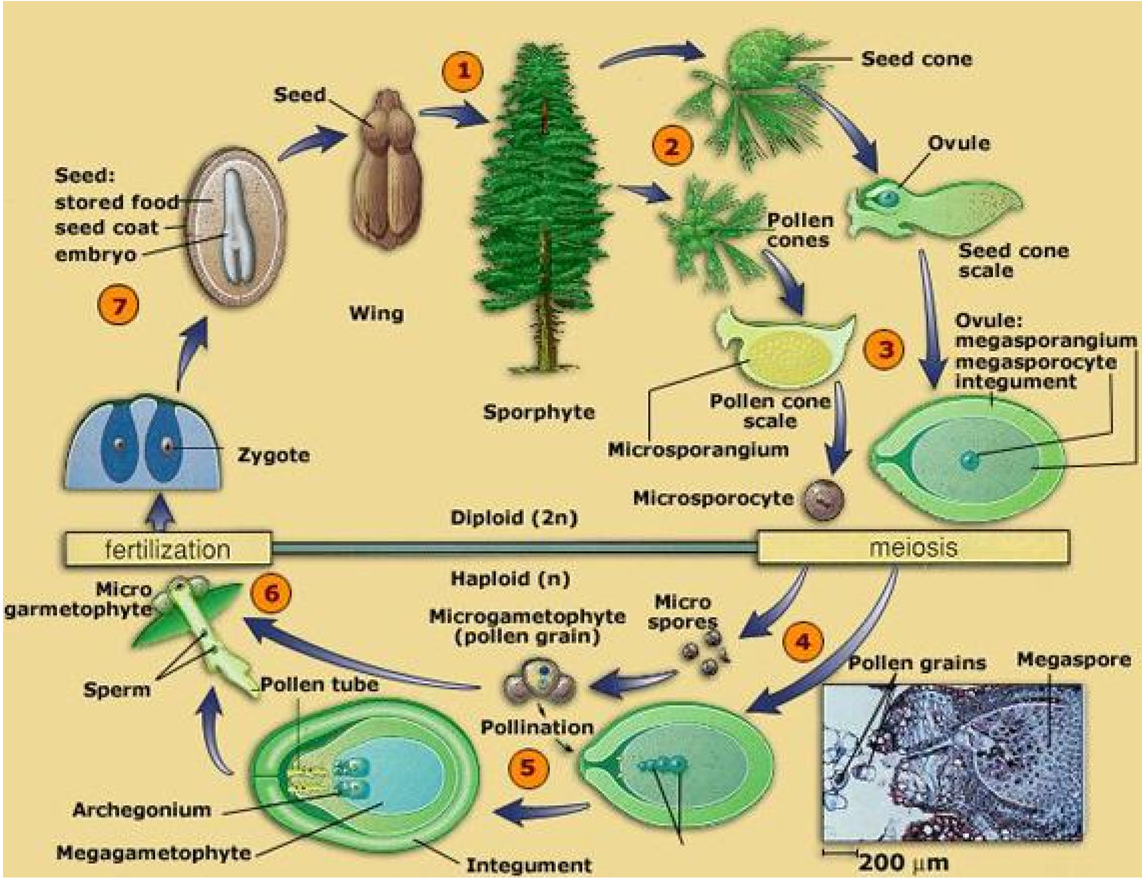

Seed cone
Seed
Ovule
Seed:
stored food
seed coat
embryo
Pollen cones
Seed cone scale
Wing
Ovule:
megasporangium
megasporocyte
integument
Sporphyte
Pollen cone scale
Microsporangium
Zygote
Microsporocyte
Diploid (2n)
fertilization
meiosis
Haploid (n)
Micro garmetophyte
Micro spores
Microgametophyte (pollen grain)
Sperm
Pollen tube
Pollen grains
Megaspore
Pollination
Archegonium
Megagametophyte
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200 µm
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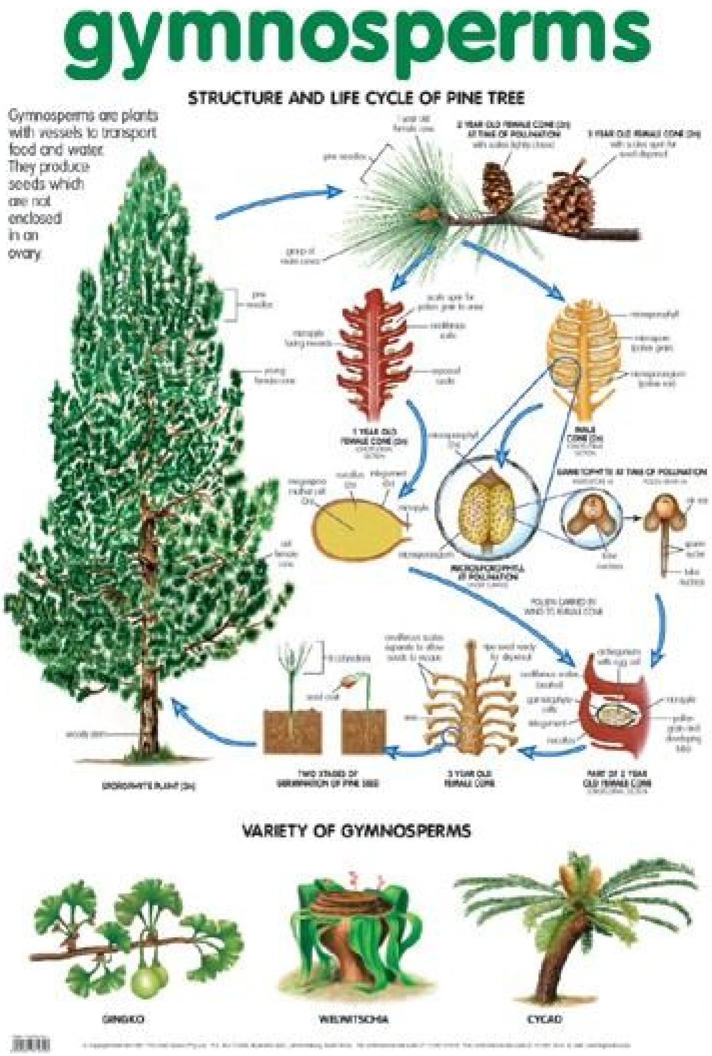


Life cycle of a gymnosperm

ANGIOSPERMS

The angiosperms, or flowering plants, are one of the major groups of extant seed plants and arguably the most diverse major extant plant group on the planet, with at least 260,000 living species classified in 453 families. They occupy every habitat on Earth except extreme environments such as the highest mountaintops, the regions immediately surrounding the poles, and the deepest oceans. They live as epiphytes (i.e., living on other plants), as floating and rooted aquatics in both freshwater and marine habitats, and as terrestrial plants that vary tremendously in size, longevity, and overall form. They can be small herbs, parasitic plants, shrubs, vines, lianas, or giant trees. There is a huge amount of diversity in chemistry (often as a defense against herbivores), reproductive morphology, and genome size and organization that is unparalleled in other members of the Plant Kingdom. Furthermore, angiosperms are crucial for human existence; the vast majority of the world's crops are angiosperms, as are most natural clothing fibers. Angiosperms are also sources for other important resources such as medicine and timber.

Characteristics

Despite their diversity, angiosperms are clearly united by shared, derived features including 1) ovules that are enclosed within a carpel, that is, a structure that is made up of an ovary, which encloses the ovules, and the stigma, a structure where pollen germination takes place, 2) double fertilization, which leads to the formation of an endosperm (a nutritive tissue within the seed that feeds the developing plant embryo), 3) stamens with two pairs of pollen sacs, 4) features of gametophyte structure and development, and 5) phloem tissue composed of sieve tubes and companion cells.

Angiosperm Longevity

The longevity of angiosperms vary widely, with some living hundreds of years -- trees, for instance -- while others die after only one season. These life cycles are measured in a circular fashion, from seed to seed. If an angiosperm lives for one growing season or one year, it is considered an annual. Those that have a life cycle of two years or growing seasons are called biennials, while those living three or more growing seasons are called perennials.

Annuals

Annuals, like all other angiosperms, sprout from a seed. If two seed leaves sprout, the annual is a dicotyledon (or dicot), the largest group of angiosperms. If the seed sprouts one leaf, it is a monocotyledon, or monocot. Vegetative growth like roots, stems and leaves continues to develop. When the plant is ready to reproduce, it flowers. Depending on the type of plant, pollination occurs within a flower, between flowers on a single plant or between plants. The process of pollination accomplishes fertilization, fusing sperm and egg. From this, a seed contained in a protective covering or fruit is produced. After this production, the plant dies.

Biennials

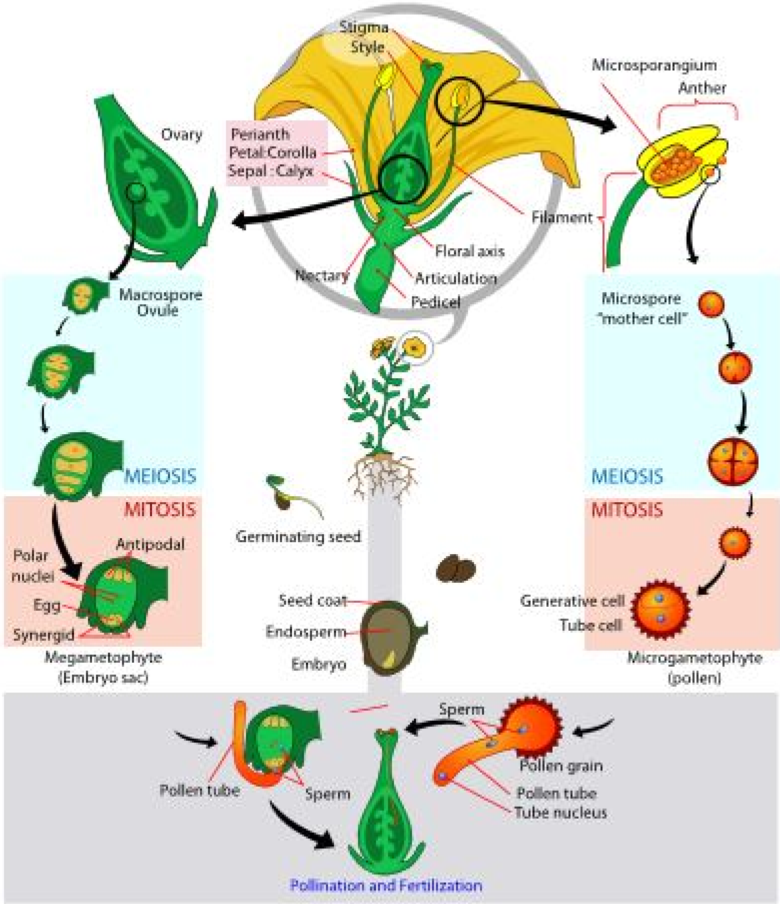
Similar to an annual, after a biennial sprouts from a seed it creates vegetative matter. Unlike the annual, though, the biennial does not flower. Instead, it creates food storage organs like bulbs or tubers such as potatoes to fuel future growth. Then, they go into a period of dormancy for winter. When the plant becomes active again in the next growing season, it has another stage of vegetative growth, then flowers, reproduces and dies

Perennials

Perennials may be woody or herbaceous. Both kinds sprout from seed, grow and flower. The stems of the herbaceous perennials, having no protective bark, die over the winter, but the roots survive; in the spring, new growth arises from that base. The woody perennials, with their bark, merely go into dormancy over the winter and renew their growth the following spring. Perennials do not necessarily flower every year. Some perennials don't flower for many years.

STEPS OF THE LIFE CYCLE OF ANGIOSPERMS

Microspores are produced during meiosis on the anther.2. Pollen grains develop from a microspore. Two sperm form through division of generative cells and the pollen tube is produced.3. Four megaspores are produced by meiosis in the megasporangium of the ovule, with one surviving to form a gametophyte.4. Two sperm cells are discharged through the pollen tube into each ovule after pollination.5. Double fertilization results in a zygote (2n) and an endosperm (3n).6. A developing zygote (embryo) and the food source are packaged into a seed.7. The embryo becomes a mature sporophyte when the seed germinates



Difference between Gymnosperms and Angiosperms

Gymnosperms	Angiosperms
A gymnosperm (Greek for "naked seed") is a vascular plant that produces seeds that are not protected by fruit but are hidden in a woody cone	Angiosperms are vascular plants that produce flowers and fruit with one or more seeds Angiosperms develop their seeds inside the ovary
They do not have flowers, but most retain their leaves year round	Angiosperms have flowers
Gymnosperms include over 600 species in four divisions: Conifers, Cycads, Ginkgoes, and Gnetophytes.	Angiosperms make up two classes: monocotyledonous (monocot) and dicotyledonous (dicot) plants
Most of today's gymnosperms belong to the conifer division. Gymnosperms are found in most of the world's regions and take most of the credit for timber and paper products	Angiosperms make up more than 80% of all plant species, ranging from roses to palm trees.
The endosperm is (n haploid) reproduction is achieved through male and female cones	The endosperm is (3n triploid) reproduction is achieved through developed female structures (anther and stigma)
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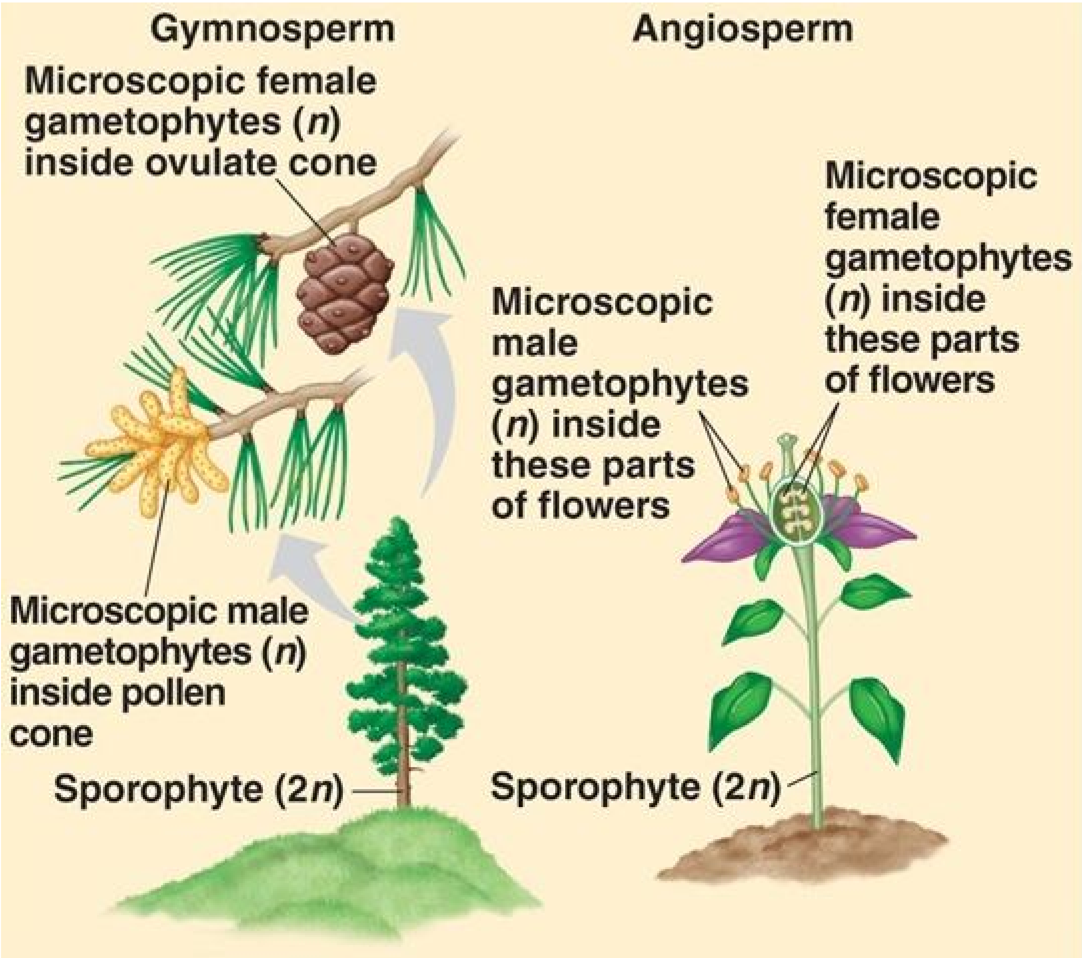

PRACTICE QUESTIONS

1. List the characteristics of the higher plants.

2. Describe the gymnosperms life cycle using a clear diagram

3 Identify the Characteristics of Gymnosperm

4. Explain the Economic Importance of Gymnosperms

5. Differentiate between angiosperms and gymnosperms

6. Write an essay on the concept of longevity in angiosperm

7. Using the angiosperm life cycle, explain explicitly the concept of double fertilization